Boda Shanmukh Datta

+91-9948511771 | shanmukhadattaboda069@gmail.com | LinkedIn | GitHub | Portfolio

Education

Dr. B.R. Ambedkar National Institute of Technology Jalandhar Jul 2024 – Jun 2028
B.Tech in Instrumentation and Control Engineering; Minor in Artificial Intelligence
CGPA: 7.95/10 | **AI Minor:** 9.2/10
Coursework: DSA, Operating Systems, Computer Networks, DBMS, OOP, ML, Probability & Statistics

Technical Skills

Languages: Python, SQL, C++ (basic)
AI/ML: PyTorch, TensorFlow/Keras, scikit-learn, LangChain, LangGraph
Backend & Web: FastAPI, Flask, REST APIs, React, Node.js, OAuth 2.0, JWT
Tools & Databases: Git, Linux, Docker, AWS, CI/CD basics, MySQL, MongoDB, SQLite
Quantum Computing: PennyLane, Qiskit

Experience

Center for Training & Placement, NIT Jalandhar Jalandhar, India
Web Development Team Member *Dec 2025 – Present*

- Develop and maintain a production MERN placement portal, including REST APIs for student records and company registration; collaborate through Git-based code review to debug and deliver features.

Infosys Springboard Remote
Software Engineering Intern – AI Music Composition System *Aug 2025 – Nov 2025*

- Built and deployed a React/Flask application integrating YAMNet audio analysis for mood detection; implemented Base64 audio transfer and IndexedDB caching to reduce redundant server requests.

QWorld – QIntern 2026 Remote
Research Intern – Quantum Machine Learning *Jun 2026 – Aug 2026*

- Built a four-class malware classification pipeline combining gradient-boosted feature selection and kernel-based classifiers; investigated generalization across CIC-MalMem-2022 and EMBER datasets.

Projects

No Hack – Network Threat Detection | *Python, XGBoost, FastAPI* GitHub

- Built an NSL-KDD intrusion-detection pipeline combining boosted classifiers with anomaly detectors; fused model agreement and disagreement signals through a stacked meta-model and served inference with FastAPI.
- Designed a Windows extension with network-flow collection, a local dashboard, explainable risk scores, and drift monitoring to distinguish known threats, novel anomalies, and uncertain predictions.

ChronoMind AI – Planning Agent | *Python, FastAPI, LangGraph, React* GitHub

- Built a personal planning application with task tracking, calendar views, voice input, Google/Firebase authentication, and a LangGraph workflow for rescheduling unfinished tasks.
- Designed an adaptive planning architecture for goal-aware prioritization, task splitting, constraint validation, and bounded replanning, with user approval for high-impact schedule changes.

DermaLens AI – Skin-Condition Analysis | *Python, TensorFlow/Keras, FastAPI* GitHub

- Designed a serving platform around an existing six-class facial skin-condition research pipeline, preserving model-specific preprocessing and compatibility with exported CNN and Vision Transformer artifacts.
- Specified image-quality checks, confidence-based abstention, out-of-distribution detection, Grad-CAM explanations, and validated feedback for controlled model updates.

VoiceDoc – Offline Health Assistant | *Planned: Kotlin, whisper.cpp, SQLite* GitHub

- Designed an Android voice-to-voice architecture combining offline speech recognition, local retrieval-augmented generation, a quantized language model, and speech synthesis for Telugu, Tamil, and English.
- Defined curated knowledge retrieval, source-grounded responses, urgent-symptom escalation, and memory-budget controls for private on-device health information; implementation and device validation planned.

Achievements & Leadership

- **Amazon ML Summer School 2026 Scholar:** Selected for advanced training in machine learning.
- **Institute Internship Representative, NIT Jalandhar:** Coordinate student internship communication.
- **AI Lead, Google Developer Groups on Campus (GDGC), NIT Jalandhar.**
- **GSSoC 2026:** Contributed to CodeGraphContext, improving cross-file call-graph resolution.